Table 1. Comparison of $^{40}Ar/^{39}Ar$ ages (Sieh et al., 2019; PKS-B and PKS-H updated, K. Sieh, written communication, 2026) with measured paleomagnetic polarity for the dated Bolaven sites, in order of age. Expected chron and polarity are those spanned by the age ± 2σ on the GPTS 2020 (Ogg, 2020; Brunhes–Matuyama boundary 773 ka; Jaramillo 990–1071 ka; Cobb Mountain 1180–1215 ka). Measured polarity is assigned from the latitude of the site-mean virtual geomagnetic pole (VGP; Table 2), not from the sign of inclination. Coordinates are means of the 2024 GPS waypoints (WGS84). Shaded rows are polarity mismatches.

Site	Sample (Sieh et al.)	Lat (°N)	Lon (°E)	Setting	$^{40}Ar/^{39}Ar$ age (ka ± 2σ)	Expected chron (GPTS 2020)	Expected polarity	VGP lat (°)	Measured polarity	Age–polarity agreement
PKS-K	PKS24-K	15.2419	106.2585	In-situ flow	407 ± 18	C1n	N	−75.3	R	Mismatch
NWF-A	NWF24-A	15.6446	106.1427	In-situ flow	518 ± 6	C1n	N	+78.1	N	Match
PKS-I	PKS24-I / PKS15-I	15.2328	106.1908	Corestone (ex situ)	553 ± 20	C1n	N	+31.3	Indet.[a]	Displaced clast
PKS-J	PKS24-J	15.2431	106.2581	In-situ flow	601 ± 44	C1n	N	−84.7	R	Mismatch
PKS-HA	PKS24-HA	15.1557	106.1918	Upper horizon	770.5 ± 9.3	C1n / C1r.1r[b]	N or R	−52.6 / −65.3	R, R (64° apart)[c]	Discordant
PKS-HB	PKS24-HB	15.1557	106.1918	Lower flow, 5.5 m below HA	> 770.5[d]	C1n / C1r.1r[b]	N or R	−55.0	R	Consistent if > 773 ka
NWF-B	NWF24-B	15.6236	105.9859	Blocks (transported)	934 ± 30	C1r.1r	R	+0.2	Indet.[a]	Random clasts (Fig. 8o)
PKS-B	PKS24-BA-1	15.1543	106.1897	In-situ flow	1155 ± 2[e]	C1r.2r	R	−75.6	R	Match
WBE-A	WBE24-A	15.2700	106.3472	Basal flow	1397 ± 13	C1r.3r	R	−78.0	R	Match

[a] VGP latitude between −45° and +45° (PKS-I) or a random clast population (NWF-B, Watson test, Fig. 8o): polarity not assigned.

[b] The 770.5 ± 9.3 ka age (761–780 ka) spans the Brunhes–Matuyama boundary at 773 ka and the ~800–770 ka reversal-transition interval.

[c] Core-level VGP latitudes of the two PKS-HA cores (HA1 / HA2), both reversed; the cores are internally tight but 64° apart (Table 2).

[d] PKS-HB is not independently dated; it underlies the dated HA horizon by 5.5 m and is therefore older than 770.5 ka.

[e] Updated age for sample PKS24-BA-1 (K. Sieh, written communication, 2026); previously reported as ~1.0 Ma.

Table 2. Sample-to-site paleomagnetic statistics for the dated Bolaven localities, computed by specimen → sample (oriented core or block) → site Fisher (1953) averaging of the high-temperature (HT, ChRM) component in geographic coordinates (no tilt correction; flows are sub-horizontal). Specimen fits are from the pmagpy re-fits (fitquest; deposited in MagIC) and are accepted where MAD ≤ 15°; n spec gives accepted/measured specimens. α95 and k are core-level statistics; for single-core sites (*) they are specimen-level. Indented italic rows give individual core means where the cores of a site disagree or are few. VGPs are computed at each site's GPS coordinates (Table 1). Polarity is from VGP latitude. Dashes denote statistics that are undefined (single core) or meaningless (α95 ≥ 90°). Shaded rows are polarity mismatches.

Site / core	Age (ka ± 2σ)	N cores (n spec acc./meas.)	Dec (°)	Inc (°)	α95 (°)	k	VGP lat (°)	VGP lon (°E)	Polarity (VGP)	Expected (GPTS)	Result
PKS-K	407 ± 18	2 (9/9)	195.1	−33.2	86.6	10.6	−75.3	2.9	R	N	Mismatch
PKSK-1		*1 (3)*	*174.9*	*−29.4*	*8.3*	*222.6*	*−85.0*				
PKSK-2		*1 (6)*	*216.3*	*−33.5*	*2.7*	*605.4*	*−55.2*				
NWF-A	518 ± 6	5 (15/15)	11.8	+23.7	11.8	42.9	+78.1	210.5	N	N	Match
PKS-I	553 ± 20	1 (5/5)	47.9	−35.6	4.2*	330.4*	+31.3	231.3	Indet.	N	Clast
PKS-J	601 ± 44	3 (12/12)	174.9	−31.9	7.9	244.2	−84.7	219.8	R	N	Mismatch
PKS-HA	770.5 ± 9.3	2 (19/19)	169.4	−10.1	–	3.3	−75.5	153.3	R	N/R	Discordant
HA1		*1 (11)*	*141.2*	*−28.0*	*4.7*	*96.4*	*−52.6*				
HA2		*1 (8)*	*194.5*	*+9.8*	*4.4*	*157.3*	*−65.3*				
PKS-HB	> 770.5	1 (5/5)	209.6	+7.9	3.3*	549.1*	−55.0	46.9	R	N/R	Coherent
NWF-B	934 ± 30	3 (20/20)	75.6	−60.2	84.8	3.2	+0.2	239.1	Indet.	R	Random (congl.)
NWFB-1		*1 (7)*	*120.5*	*−41.5*	*2.8*	*466.3*	*−33.8*				
NWFB-2		*1 (6)*	*17.8*	*−26.3*	*0.9*	*6061.9*	*+55.7*				
NWFB-3		*1 (7)*	*128.4*	*−73.2*	*2.2*	*785.2*	*−32.7*				
PKS-B	1155 ± 2	1 (7/7)	190.6	−10.6	3.7*	269.8*	−75.6	58.3	R	R	Match
WBE-A	1397 ± 13	2 (11/11)	168.1	−34.5	28.0	81.7	−78.0	216.1	R	R	Match
WBEA-1		*1 (8)*	*163.0*	*−39.4*	*2.0*	*779.6*	*−72.4*				
WBEA-2		*1 (3)*	*172.6*	*−29.5*	*3.4*	*1285.1*	*−82.9*				

Sites in order of age. PKS-H is reported as its two stratigraphic horizons (HA upper, dated; HB 5.5 m below). PKS-I and NWF-B are included for completeness: PKS-I is an ex-situ corestone and NWF-B a random clast population (conglomerate test, Fig. 8o).